

chapter 3

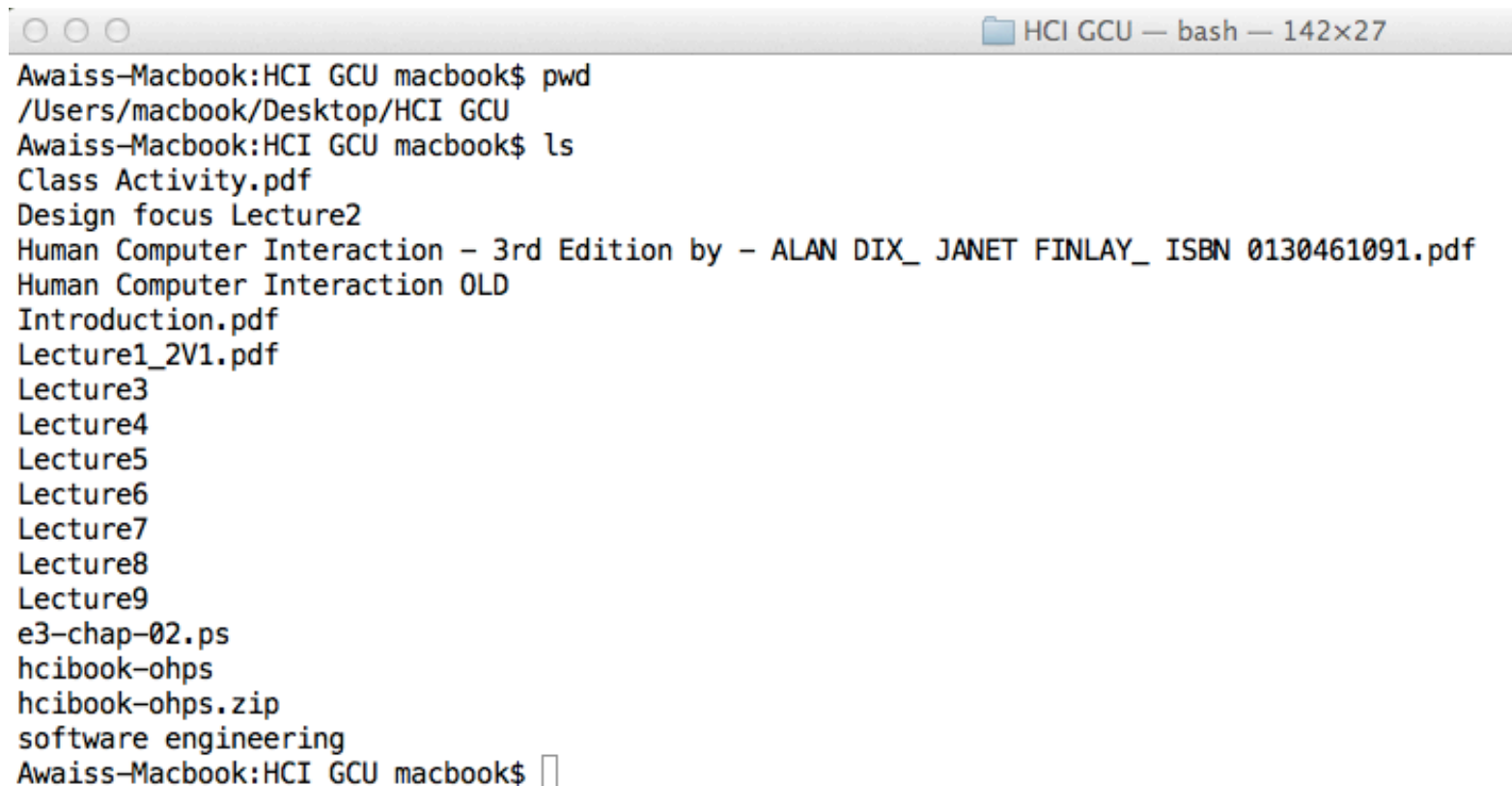
the interaction

INTERACTION STYLES

- The choice of interface style can have a profound effect on the efficiency of interaction between humans and computers.
- We will discuss the most common interface styles and note the different effects these have on the interaction. There are a number of common interface styles including:
 - Command line interface
 - Menus
 - Natural language
 - Question/answer
 - Form-fills and spreadsheets
 - WIMP
 - Point and click
 - Three-dimensional interfaces.

Command line interface

- The command line interface was the first interactive dialog style to be commonly used and, in spite of the availability of menu-driven interfaces, it is still widely used. WHY?

A screenshot of a terminal window titled "HCI GCU — bash — 142x27". The window shows a user at the "Awaiss-Macbook:HCI GCU macbook\$ prompt. The user has entered the command "pwd" and received the output "/Users/macbook/Desktop/HCI GCU". Then, the user entered "ls" and received a list of files and directories: "Class Activity.pdf", "Design focus Lecture2", "Human Computer Interaction – 3rd Edition by – ALAN DIX_ JANET FINLAY_ ISBN 0130461091.pdf", "Human Computer Interaction OLD", "Introduction.pdf", "Lecture1_2V1.pdf", "Lecture3", "Lecture4", "Lecture5", "Lecture6", "Lecture7", "Lecture8", "Lecture9", "e3-chap-02.ps", "hcibook-ohps", "hcibook-ohps.zip", "software engineering". The prompt is now "Awaiss-Macbook:HCI GCU macbook\$ " with a cursor.

```
Awaiss-Macbook:HCI GCU macbook$ pwd
/Users/macbook/Desktop/HCI GCU
Awaiss-Macbook:HCI GCU macbook$ ls
Class Activity.pdf
Design focus Lecture2
Human Computer Interaction – 3rd Edition by – ALAN DIX_ JANET FINLAY_ ISBN 0130461091.pdf
Human Computer Interaction OLD
Introduction.pdf
Lecture1_2V1.pdf
Lecture3
Lecture4
Lecture5
Lecture6
Lecture7
Lecture8
Lecture9
e3-chap-02.ps
hcibook-ohps
hcibook-ohps.zip
software engineering
Awaiss-Macbook:HCI GCU macbook$
```

Command line interface

- Provides a means of expressing instructions to the computer directly, using function keys, single characters, abbreviations or whole-word commands.
- In some systems (legacy systems now) the command line is the only way of communicating e.g MS-DOS.
- *BUT* more commonly today it is supplementary to menu-based interfaces, providing fast access to the system's functionality for experienced users.

Pros:

- Command line interfaces are powerful in that they offer direct access to system functionality (as opposed to the hierarchical nature of menus)
- Can be combined to apply a number of tools to the same data. For example execute a script to convert all file names in a directory to small case and remove special characters (used in Android).
- Often has a number of options or parameters that will vary its behavior in some way, and it can be applied to many objects at once, making it useful for repetitive tasks.

Cons:

- Difficulty in use and learning. Commands must be remembered, as no cue is provided in the command line to indicate which command is needed.
- Better for expert users than for novices.
- This problem can be alleviated a little by using consistent and meaningful commands and abbreviations. The commands used should be terms within the vocabulary of the user rather than the technician. Unfortunately, commands are often obscure and vary across systems(Git/SVN/BitBucket), causing confusion to the user and increasing the overhead of learning.

Menus

- In a menu-driven interface, the set of options available to the user is displayed on the screen, and selected using the mouse, or numeric or alphabetic keys.
- So it supports recognition or recall?

PAYMENT DETAILS

P3-7

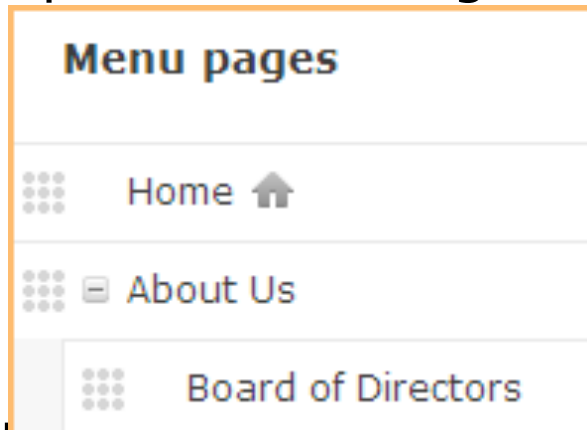
please select payment method:

1. cash
2. check
3. credit card
4. invoice

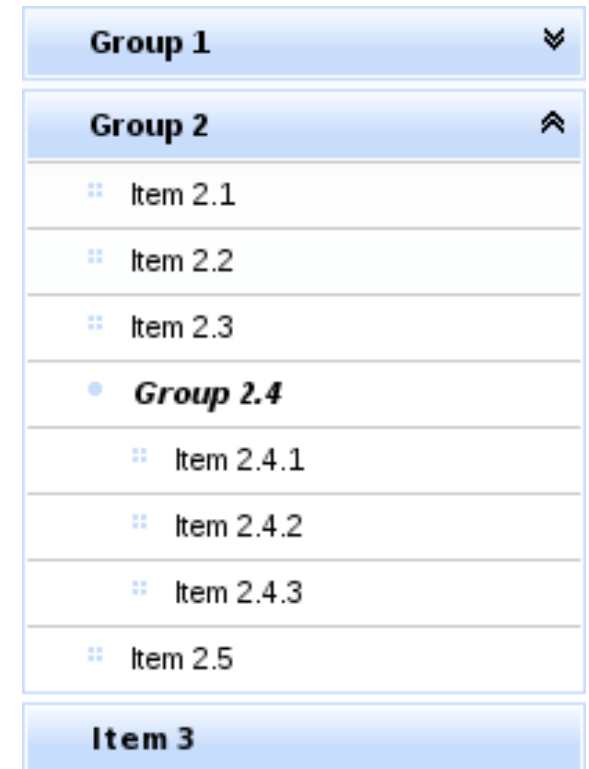
9. abort transaction

Pros and Cons:

- Menu options need to be meaningful and logically grouped to aid recognition.



- Often menus are hierarchically ordered(user has to click more than once to navigate a tree) and the option required is not available at the top layer of the hierarchy. The grouping and naming of menu options then provides the only cue for the user to find the required option.



Pros and Cons:

- Such systems either can be purely text based, with the menu options being presented as numbered choices or may have a graphical component in which the menu appears within a rectangular box and choices are made, perhaps by typing the initial letter of the desired selection(drop down menu), or by entering the associated number, or by moving around the menu with the arrow keys.

Natural language

- Perhaps the most attractive means of communicating with computers.
- It would be great if Users, unable to remember a command or lost in a hierarchy of menus, would be able to give commands to computer that expressed in everyday words BUT still a lot of limitations.

Pros and Cons:

- Ambiguity of natural language makes it very difficult for a machine to understand.
- Language is ambiguous at a number of levels(semantic vs syntactic).
- First, the syntax, or structure, of a phrase may not be clear. If we are given the sentence
The boy hit the dog with the stick
we cannot be sure whether the boy is using the stick to hit the dog or whether the dog is holding the stick when it is hit.
- Even if a sentence's structure is clear, we may find ambiguity in the meaning of the words used. For example, the word 'pitch' may refer to a sports field, a throw, sound quality etc.

Pros and Cons:

-Humans use context and general knowledge to sort out these ambiguities. This information is difficult to provide to the machine.

-However, systems can be built to understand restricted subsets of a language. For a known and constrained domain, the system can be provided with sufficient information to disambiguate terms

-Use of natural language in restricted domains is relatively successful, but user still has to learn which phrases the computer understands and may become frustrated if too much is expected.

Question/answer

-Question/answer dialog is a simple mechanism for providing input to an application in a specific domain. The user is asked a series of questions (mainly with yes/no responses, multiple choice, or codes) and so is led through the interaction step by step. An example of this would be **web questionnaires**.

Pros and Cons:

-These interfaces are easy to learn and use, but are limited in functionality and power. As such, they are appropriate for restricted domains.

Questionnaire Review

Hold Name Aaron Stinson Wrongful Termination

Open Date 07/15/2010

Your E-mail Address *

Employment Questions

- * 1. What is your date of employment with ABC Company?

- * 2. Are you a full-time employee of ABC Company?

☐ Yes
☐ No

- * 3. Did you receive an employee handbook?

☐ Yes
☐ No

Department Questions

- * 4. What is your employment date for your current department?

- * 5. Have you worked in the Engineering Department at any time during your employment?

☐ Yes
☐ No

Legal Hold Specific Questions

- * 6. Did you receive any e-mails from Bryan Calhoun between May 1, 2009, and July 31, 2009?

☐ Yes
☐ No

- * 7. Were you present at the Engineering Department meeting on June 27, 2009?

☐ Yes
☐ No

If you answered "Yes" to any Legal Hold Specific Questions above, please fill out the information below:

Data Collection Questions

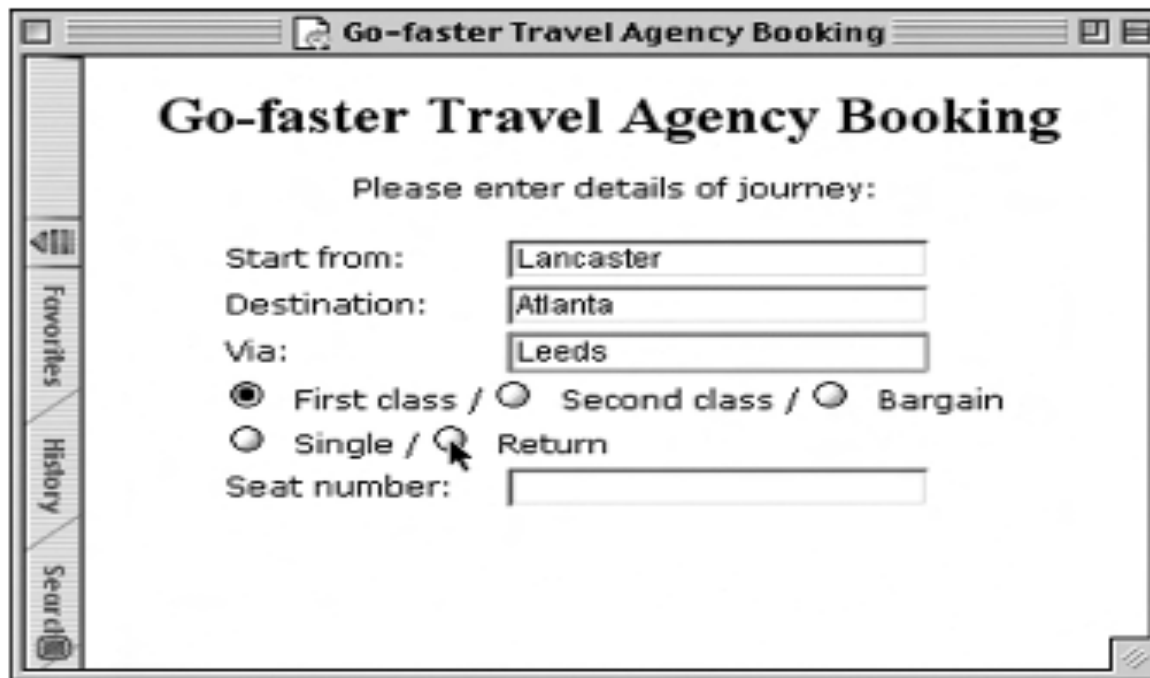
The following questions will aid the Legal Department in collecting any data that you have that is relevant and responsive to the request for discovery. Please answer all questions below.

8. Do you have a company-issued laptop?

☐ Yes
☐ No

Form-fills and spreadsheets

-Form-filling interfaces are used primarily for data. The user is presented with a display resembling a paper form, with slots to fill in. Often the form display is based upon an actual form with which the user is familiar, which makes the interface easier to use.



The screenshot shows a web browser window with the title "Go-faster Travel Agency Booking". The main heading inside the window is "Go-faster Travel Agency Booking". Below the heading, it says "Please enter details of journey:". The form contains the following fields and options:

- Start from: Lancaster
- Destination: Atlanta
- Via: Leeds
- Class: ☒ First class / ☐ Second class / ☐ Bargain
- Single / ☒ Return
- Seat number:

On the left side of the window, there is a vertical menu with the following options: Favorites, History, and Search.

Pros and Cons:

- Most form-filling interfaces allow easy movement around the form and allow some fields to be left blank (non compulsory).
- Spreadsheets** are a sophisticated variation of form filling. The spreadsheet comprises a grid of cells, each of which can contain a value or a formula.
- The user can therefore manipulate values to see the effects of changing different parameters. Spreadsheets are an attractive medium for interaction: the user is free to manipulate values at will, making the interface more flexible and natural.

Form-fills and spreadsheets cntd

	Pooches Pet Emporium				
Date	Description	Dog	Income	Outgoings	Balance
9/2/02	Fees – Mr C. Brown	Snoopy	96.37		96.37
10/2/02	Rubber bones			36.26	60.11
10/2/02	Fees – Mrs E. R. Windsor	7 corgis	1006.45		1066.56
12/2/02	Special order: 7 red carpets			47.28	992.28
16/2/02	Fees – Master T. Tin	Snowy	32.98		1025.26
17/2/02	Beefy Bruno's Bonemeal			243.47	781.79
21/2/02	Fees – Mr F. Flintstone	Dino	21.95		803.74
21/2/02	Special order: 1 Brontosaurus bone			6.47	797.27
28/2/02	Wages – Mr S. H. Ovelit			489.46	307.81



The WIMP interface

-Currently many common environments for interactive computing are examples of the *WIMP* interface style, often simply called windowing systems.

-WIMP stands for **windows, icons, menus and pointers** and is the default interface style for the majority of interactive computer systems in use today. Examples of WIMP interfaces include Microsoft Windows, MacOS for Apple Macintosh compatibles etc

Point-and-click interfaces

-In most multimedia systems and in web browsers, virtually all actions take only a single click of the mouse button. We may point at a city on a map and when you click a window opens, showing you tourist information about the city. You may point at a word in some text and when you click you see a definition of the word.

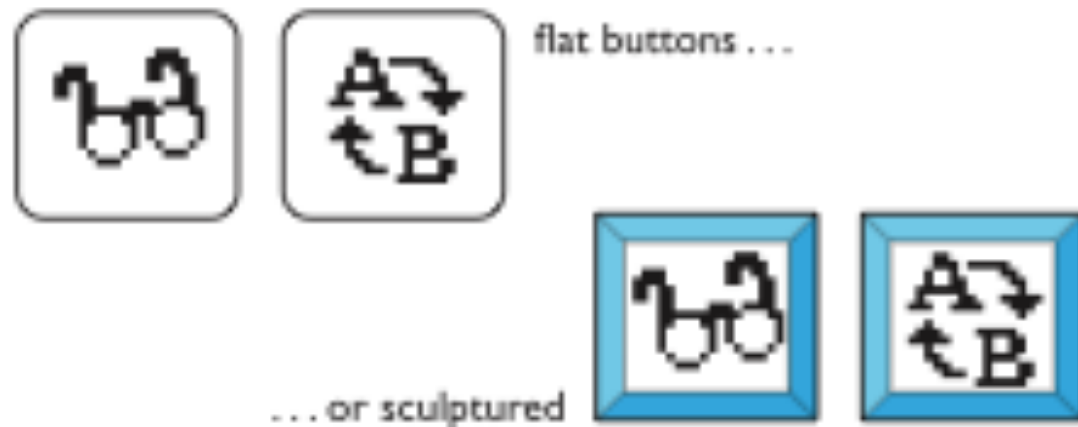
-This point-and-click interface style is obviously closely related to the WIMP style. However, the philosophy is simpler and more closely tied to ideas of ***hypertext***(presentation of information as a linked network of nodes which readers are free to navigate in a non-linear fashion).

Point-and-click interfaces cntd

- point-and-click is also extensively used in touchscreen information systems. In this case, it is often combined with a menu-driven interface.
- The point-and-click style has been popularized by world wide web pages, which incorporate all the above types of point-and-click navigation: highlighted words, maps and iconic buttons.

Three-dimensional interfaces

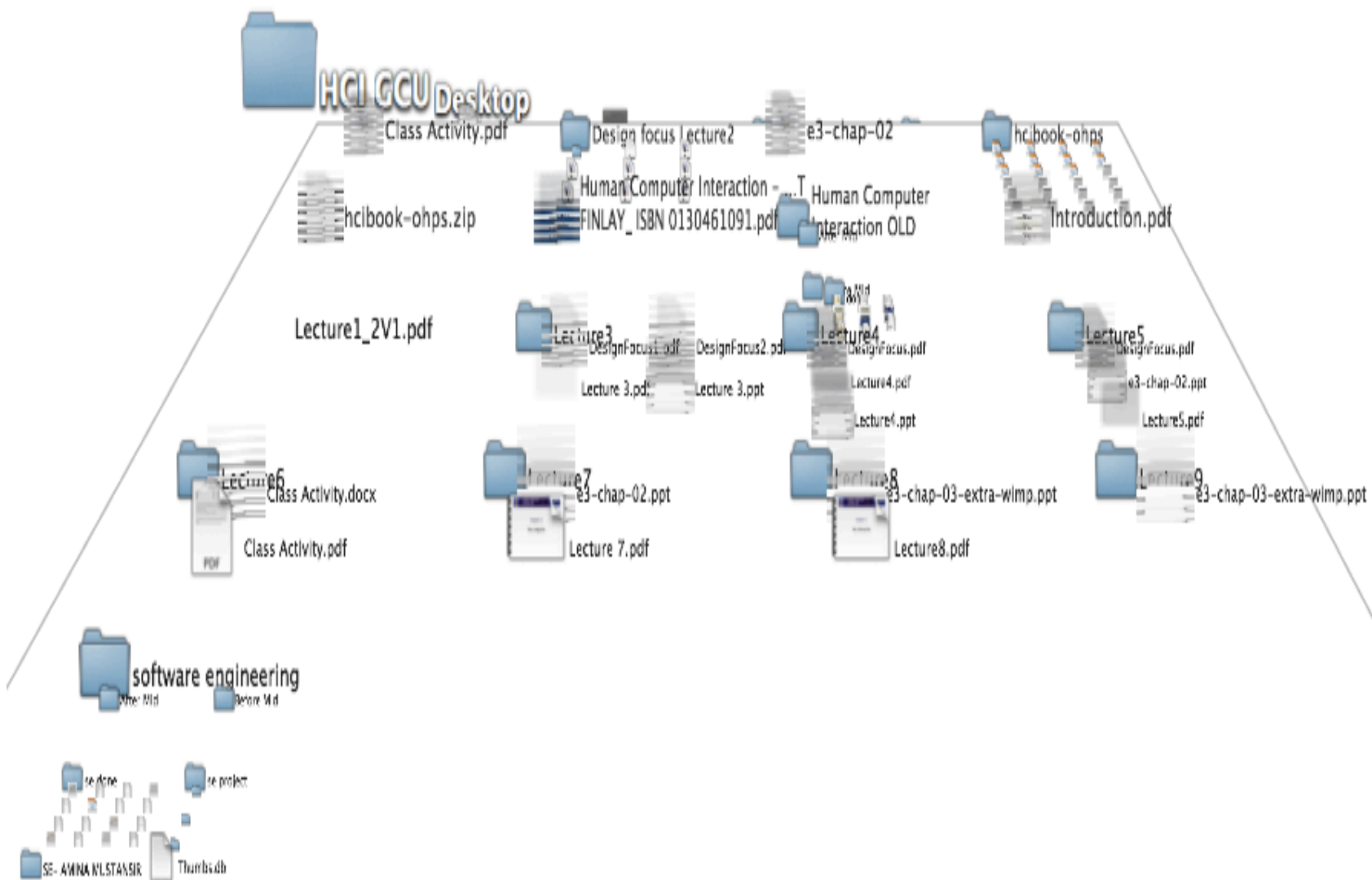
- There is an increasing use of three-dimensional effects in user interfaces.
- Simplest technique is where ordinary WIMP elements, buttons, scroll bars, etc., are given a 3D appearance using **shading**, giving the appearance of being sculpted out of stone.

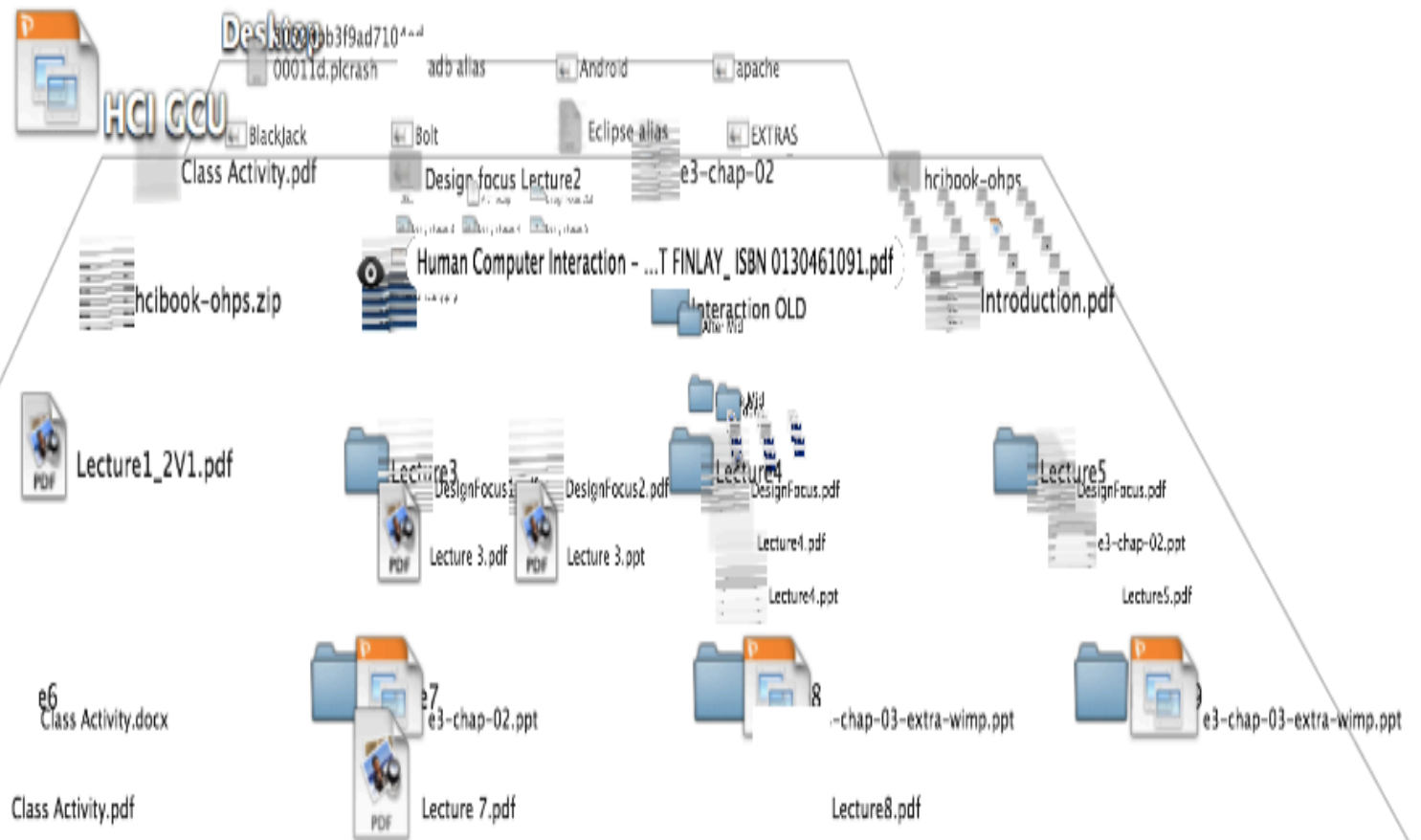


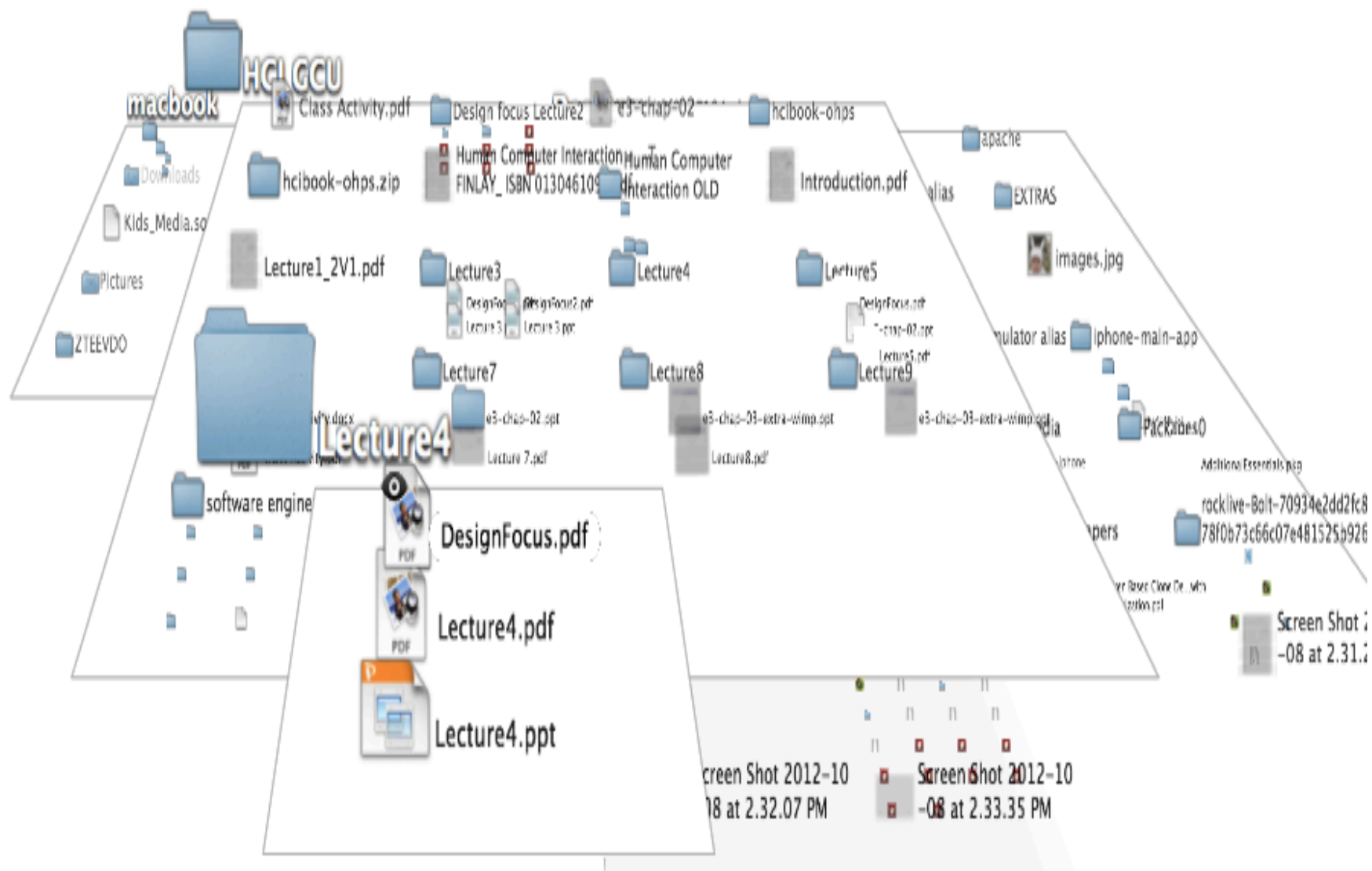
Three-dimensional interfaces



-In more complex technique the objects displayed in such systems are usually flat, but are displayed in perspective to give 3D effect. Example **3D FileSpace for MAC**







Three-dimensional interfaces

-*Many available for windows too e.g* Tactile3D - Windows program that allows users to organize files and directories in a 3D space. Users get the impression of exploring a vast virtual landscape, with each directory being a new world in which they can roam around and place objects where they want.



From all the interfaces studied so far which interface will dominate the future products in general?